Kingdom of their own

For most of the history of biology, fungi were considered to be plants. Even the great classifier of organisms Carl Linnaeus included them in his kingdom Plantae. It was only with the invention of more powerful microscopes that the differences in fungi began to be better understood. It is now known that chitin, a complex carbohydrate and component of fungus cell walls, is not found in plants. Also, fungi make their food by digesting rotted material, whereas plants make food by absorbing light energy in photosynthesis.

DNA analysis shows that fungi are far removed from plants in the evolutionary tree of life: they are, in fact, genetically closer to the branch that gives rise to animals. These same studies show that certain aquatic molds—traditionally classified as fungi—are not related to fungi, while some disease-causing microbes are fungi that have evolved to become microscopic parasites.